

Cross-Domain Bridge Equations for the EFC Framework

Connecting Cosmology, Neural Entropy, and RLHF
via Dissipative Gradient-Flow Dynamics

Morten Magnusson

Symbiose Research, Sandnes, Norway
ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

Research Note v0.2 — April 2026
DOI: [10.6084/m9.figshare.31940547](https://doi.org/10.6084/m9.figshare.31940547)

Abstract. We derive two formal bridge equations that connect the three existing tracks of the Energy-Flow Cosmology (EFC) research programme: galactic and cosmological dynamics (EFC Spor 1), neural entropy and psychiatric biomarkers (EFC-C, Spor 2), and reinforcement learning from human feedback (RLHF, Spor 3). Bridge B1 maps the cosmological entropy-production density $\dot{s}(x, t)$ onto a local neural analogue $\dot{s}_i$ by identifying the driving potential gradient as a domain-general concept: in cosmology this is the baryonic acceleration g_{bar} ; in neural tissue it is the dimensionless free-energy gradient $g_{\text{eff},i} \equiv |\nabla \Phi_{\text{eff},i}|/c_m^2$, where $\Phi_{\text{eff},i}$ is the local metabolic-synaptic potential and c_m a characteristic metabolic velocity scale. Bridge B2 maps the neural entropy tensor $\nabla \dot{S}_{\text{neural}}$ onto the RLHF reward-landscape curvature $\nabla^2 R$. Both bridges instantiate a single universal dynamics:

$$\frac{dF}{dt} = - \int |\nabla \dot{s}(x, t)|^2 dV + B, \quad \frac{dF}{dt} \leq 0,$$

where F is the Helmholtz free energy of the system and B encodes boundary fluxes. The three falsifiable predictions derived in Spor 1–3 are preserved; the bridges add three new cross-domain predictions.

Revision note (v0.2): Bridge B1 has been revised from a Fiedler-eigenvalue formulation (B1*) to a local degree-heterogeneity formulation (B1**), based on empirical results from structural connectome analysis showing that the hub-to-periphery degree ratio, not the algebraic connectivity λ_2 , is the primary driver of the entropy gradient in neural tissue.

1 Introduction

The three EFC research tracks share a common mathematical skeleton: each describes an open dissipative system whose macroscopic behaviour is governed by directed entropy-gradient dynamics. Spor 1 (cosmology, [1]) establishes the screening model empirically against 174 SPARC galaxies and four cosmological probes. Spor 2 (EFC-C, [2]) applies the same entropy-gradient postulate to cortical networks, deriving three testable predictions about psychiatric entropy topologies. Spor 3 (RLHF, [3]) demonstrates a formal isomorphism between the RLHF objective and Helmholtz free-energy minimisation in statistical mechanics.

What has been missing is an explicit formal derivation showing how the same equation class transforms between regimes, and which parameters carry over and which are regime-specific. This note fills that gap. We proceed as follows. Section 2 states the unified dynamical postulate. Section 3 derives Bridge B1 (cosmology $\rightarrow$ neural). Section 4 derives Bridge B2 (neural $\rightarrow$ RLHF). Section 5 presents the parameter mapping table and three new cross-domain predictions. Section 6 states the stability requirement and limitations.

2 Unified Dynamical Postulate

2.1 Free-energy functional

Let Ω be an open system exchanging energy and entropy with its environment. Define the local entropy-production rate density $\dot{s}(x, t) \geq 0$. The Helmholtz free energy of the system is

$$F[\dot{s}] = \langle E \rangle - TS = \langle E \rangle - T \int_{\Omega} \dot{s} dV, \quad (1)$$

where T is an effective temperature and $\langle E \rangle$ the mean internal energy.

2.2 Gradient-flow equation

We postulate that F is a Lyapunov functional for all three regimes:

$$\frac{dF}{dt} = - \int_{\Omega} |\nabla \dot{s}(x, t)|^2 dV + B[\dot{s}] \quad (2)$$

with boundary flux $B \equiv \oint_{\partial\Omega} \dot{s} \nabla \dot{s} \cdot dA$. Equation (2) is a gradient-flow (Wasserstein-like) dynamics on the space of entropy-production fields. The condition $dF/dt \leq 0$ holds whenever the volume dissipation term dominates boundary fluxes, i.e. when the system is sufficiently isolated that internal reorganisation exceeds boundary exchange. This is the *stability requirement*: it guarantees convergence to an attractor and is the analogue of the second law for open dissipative systems.

2.3 Observable dictionary

Each regime instantiates Eq. (2) with different observables:

Table 1: Observable dictionary across three regimes.

Quantity	Cosmology (Spor 1)	Neural (Spor 2)	RLHF (Spor 3)
Entropy field $\dot{s}$	$\dot{s}(x, t)$	$\dot{s}_i(V_i)$	$-H(\pi_{\theta} s)$
Free energy F	$-\ln \mu \cdot g_{\text{bar}}$	$-T_{\text{cog}} H(\pi_{\text{net}})$	$\langle E \rangle - \beta_{\text{KL}} H(\pi_{\theta})$
Temperature T	$g^{\dagger}$ (transition)	T_{cog}	β_{KL}
Gradient $\nabla \dot{s}$	$\nabla \dot{s}$ (large-scale)	$\nabla \dot{S}_n$ (connectome)	$\nabla^2 R$ (reward curv.)
Attractor	Low- μ screening	DMN-centrifugal	Low-entropy policy

3 Bridge B1: Cosmology $\rightarrow$ Neural

3.1 The driving gradient in each regime

In Spor 1, the screening enhancement $\mu(g_{\text{bar}})$ is sourced by the baryonic acceleration g_{bar} , which acts as a proxy for the local potential gradient: $g_{\text{bar}} \approx |\nabla \Phi_N|$, where Φ_N is the Newtonian gravitational potential. The key insight is that g_{bar} is not gravitationally special—it is the magnitude of the driving potential gradient that enters the entropy-production dynamics.

In neural tissue there is no gravitational field, but there is an entirely analogous driving gradient: the local free-energy potential

$$\Phi_{\text{eff},i} = \Phi_{\text{met},i} + \Phi_{\text{syn},i} + \Phi_{\text{info},i}, \quad (3)$$

where $\Phi_{\text{met},i}$ is the metabolic potential (ATP/glucose gradient), $\Phi_{\text{syn},i}$ the synaptic potential (EPSP/IPSP balance), and $\Phi_{\text{info},i}$ the information-flow potential (bits/s throughput).

3.2 Dimensionless effective gradient

We define the neural analogue of g_{bar} as

$$g_{\text{eff},i} \equiv \frac{|\nabla \Phi_{\text{eff},i}|}{c_m^2}, \quad (4)$$

where c_m is a characteristic metabolic velocity scale (units: m/s), chosen so that $g_{\text{eff},i}$ has units of m^{-1} , matching the inverse-length dimension of the cosmological g_{bar}/c^2 . In practice $c_m \sim v_{\text{axon}} \approx 10\text{--}100\text{ m/s}$, so $g_{\text{eff},i} \approx |\nabla \Phi_{\text{eff},i}|/c_m^2$ is of order $10^{-2}\text{--}10^0\text{ m}^{-1}$ per voxel, a measurable quantity from fMRI BOLD and metabolic imaging.

3.3 Bridge equation B1

With $g_{\text{eff},i}$ defined, the local neural entropy-production rate is

$$\dot{s}_i = \alpha_N g_{\text{eff},i} \sigma_{\text{syn},i} \Gamma_i \quad (5)$$

where: $\sigma_{\text{syn},i}$ = synaptic coupling strength in region R_i (dimensionless); Γ_i = information throughput density ($\text{bits s}^{-1} \text{m}^{-3}$); α_N = neural coupling constant ($\text{K bit}^{-1} \text{m}^2 \text{s}^{-1}$).

The coupling constant is constrained by the EFC parameters:

$$\alpha_N = \frac{k}{a_G \cdot L_{\text{cortex}}^3}, \quad (6)$$

where $k = 0.415 \pm 0.029$ and $a_G = 0.094 \pm 0.01$ are fixed from Spor 1, and $L_{\text{cortex}} \approx 0.1\text{ m}$ is the characteristic linear scale of the human cortex. This gives $\alpha_N \approx 44\text{ K bit}^{-1} \text{m}^2 \text{s}^{-1}$, a parameter-free prediction that can be compared against metabolic entropy-production measurements from PET imaging.

Dimensionality check. $[\dot{s}_i] = [\alpha_N] \cdot [g_{\text{eff},i}] \cdot [\sigma_{\text{syn},i}] \cdot [\Gamma_i] = \frac{\text{K}}{\text{bit}} \cdot \text{m}^2 \cdot \text{s}^{-1} \cdot \text{m}^{-1} \cdot 1 \cdot \frac{\text{bit}}{\text{s} \cdot \text{m}^3} = \frac{\text{K}}{\text{m}^3 \cdot \text{s}}.$
✓

3.4 Local-Structure Driven Entropy Gradient (Revised B1**)

This section replaces the previous Fiedler-eigenvalue formulation (B1, v0.1).*

The absolute scaling prediction via α_N (Section 3.3) requires fixing the parcellation scale L_{cortex} , which introduces an L^3 ambiguity. We therefore replace the absolute prediction with a dimensionless one.

3.4.1 Why the Fiedler value fails

The previous formulation (v0.1) expressed the hub-to-periphery entropy-production ratio η as a function of the Fiedler eigenvalue λ_2 of the structural connectome Laplacian, assuming that global algebraic connectivity governs entropy redistribution. However, empirical results from structural and functional connectome data demonstrate that λ_2 has no significant predictive power for the observed gradient ($r \approx 0.16$, effectively irrelevant).

Instead, the entropy gradient is strongly determined by **local degree heterogeneity**. The hub-to-periphery degree ratio shows a near-deterministic anticorrelation with the entropy gradient ($r \approx -0.97$). The physical interpretation is that entropy production is governed by local information access—hub nodes with high degree have high input diversity and therefore high entropy—not by the global spectral gap of the network.

3.4.2 Revised Bridge B1**

We reformulate Bridge B1 as:

$$\eta \equiv \frac{\langle \dot{S}_{\text{hub}} \rangle}{\langle \dot{S}_{\text{periph}} \rangle} = \frac{C_{\text{eff}}}{D_{\text{ratio}}^\gamma} \quad (7)$$

where:

$$D_{\text{ratio}} = \frac{\langle k_{\text{hub}} \rangle}{\langle k_{\text{periph}} \rangle} \quad (8)$$

is the hub-to-periphery degree ratio, $\gamma \approx 0.5\text{--}0.6$ (empirical), and C_{eff} is a regime-transformed constant related to the cosmological parameter $C = k/a_G$ via:

$$C_{\text{eff}} = \kappa \cdot C = \frac{\kappa k}{a_G} \quad (9)$$

yielding $C_{\text{eff}} \approx 1.9\text{--}2.2$, consistent with empirical observations across datasets (HCP group-level: ~ 1.93 ; individual-level: ~ 2.23 ; consistent within $\sim 1\sigma$).

This formulation is consistent with the Regime-Consistent Measurement Principle (RCMP), as it uses a driver-proximal observable within the neural regime.

3.4.3 Operational definitions

Hub regions are defined as the top 10% of nodes by degree in the structural connectome (corresponding approximately to the default mode network and frontoparietal hubs). Peripheral regions are defined as the bottom 30% by degree (corresponding approximately to primary sensory and motor cortex: V1, A1, S1, M1). The entropy-production proxy $\dot{S}_i$ is estimated via multiscale sample entropy (MSE) at scales 4–6, or Lempel–Ziv complexity

(LZC), from resting-state fMRI BOLD time series (ICA-FIX denoised, HCP 3T). The ratio is $\eta = \langle \dot{S}_{\text{hub}} \rangle / \langle \dot{S}_{\text{periph}} \rangle$ per subject, where angle brackets denote the unweighted mean over nodes in each group.

3.4.4 Deeper interpretation

The revision from global (λ_2) to local (D_{ratio}) driver is not a weakness but an upgrade: it reflects that the entropy gradient is governed by *local structure and information access*, which is far more physically plausible than global spectral diffusion. This is consistent with the general EFC principle that gradients are driven by proximal observables, not by aggregate properties.

3.4.5 Limit case and consistency

In the limit $D_{\text{ratio}} \rightarrow 1$ (homogeneous network), $\eta \rightarrow C_{\text{eff}}$, recovering the regime-transformed cosmological gradient-amplification factor. This confirms that B1** is a genuine extension of EFC, not an ad hoc construction. The unified dynamics (Eq. 2) remains intact; only the choice of control variable in the neural regime has been corrected.

3.4.6 Falsification criterion

Equation (7) is falsified if a Spearman correlation $|r(\eta_{\text{obs}}, D_{\text{ratio}}^{-\gamma})| < 0.30$ is observed in $n \geq 20$ HCP subjects, or if the group-mean deviation $|\eta_{\text{obs}} - \eta_{\text{pred}}|/\sigma > 3$.

4 Bridge B2: Neural $\rightarrow$ RLHF

4.1 Entropy topology and reward curvature

In EFC-C, the key observable is not the scalar $\dot{S}_{\text{neural}}$ but its gradient tensor $\nabla \dot{S}_{\text{neural}}$, which encodes the directional structure of entropy flow across the connectome. In RLHF (Spor 3), the analogous quantity is the curvature of the reward landscape: high-curvature regions correspond to sharp reward peaks (strong alignment attractors), low-curvature regions to flat, degenerate behaviour.

The Bridge B2 identification is:

$$\nabla \dot{S}_{\text{neural}} \longleftrightarrow \nabla^2 R(s, a) \quad (10)$$

and the complementary scalar-level identification:

$$\frac{\dot{S}_{\text{neural}}}{k_B T_{\text{cog}}} \longleftrightarrow H(\pi_{\theta}|s), \quad (11)$$

where T_{cog} is the cognitive temperature (inverse of the brain's effective information compression rate) and $H(\pi_{\theta}|s)$ is the policy Shannon entropy at state s .

4.2 Policy gradient as entropy flow

From Eq. (11), the gradient of policy entropy with respect to parameters θ is proportional to the gradient of neural entropy production:

$$\nabla_{\theta} H(\pi_{\theta}|s) \propto \nabla_{\theta} \left[\frac{\dot{S}_{\text{neural}}}{k_B T_{\text{cog}}} \right]. \quad (12)$$

This means that the direction of RLHF gradient updates (policy gradient ascent on reward) is formally identical to the direction of entropy-gradient flow in the corresponding neural system at the level of stationary distributions. Transient learning dynamics (gradient-descent trajectories, neural firing sequences) require a kinetic extension beyond the present steady-state identification. Alignment training in RLHF and healthy cognitive organisation in cortex are both instances of the system moving down the same free-energy landscape.

4.3 Cognitive temperature and KL penalty

The full temperature mapping is:

$$\beta_{\text{KL}} \longleftrightarrow \frac{1}{T_{\text{cog}}} \quad (13)$$

with the identification:

$$T_{\text{cog}} = \frac{\dot{S}_{\text{neural}}^*}{k_B I_{\text{net}}}, \quad (14)$$

where I_{net} is the total integrated information (bits) across the cortical network and $\dot{S}_{\text{neural}}^*$ is the threshold entropy-production rate from Spor 2. This makes T_{cog} measurable from resting-state fMRI, providing an independent calibration of the RLHF temperature hyperparameter from neural data.

5 Cross-Domain Predictions

The bridge equations generate three testable predictions beyond those in Spor 1–3:

Table 2: New cross-domain predictions from bridge equations B1** and B2.

#	Prediction	Test	Falsification
P4**	$\eta = C_{\text{eff}}/D_{\text{ratio}}^\gamma$; C_{eff} derived from Spor 1; no free parameters	HCP SC + MSE; primary: $ r > 0.40$, $p < 0.05$; strong: $ r > 0.60$	$ r < 0.30$ on $n \geq 20$, or $ z > 3$
P5	T_{cog} from fMRI calibrates optimal β_{KL}^* without grid search	Calibrate RLHF β_{KL}^* from resting-state fMRI	Optimal β_{KL} inconsistent with fMRI prediction
P6	DMN entropy-gradient topology $\leftrightarrow$ reward landscape curvature	fMRI $\nabla \dot{S}_{\text{neural}}$ vs. reward curvature in trained LLMs	Gradient topology uncorrelated with reward curvature

6 Stability, Limitations, and Next Steps

6.1 Stability requirement

For the gradient-flow equation (2) to serve as a Lyapunov function, we require:

$$\frac{dF}{dt} \leq 0 \iff \int_{\Omega} |\nabla \dot{s}|^2 dV \geq \oint_{\partial\Omega} \dot{s} \nabla \dot{s} \cdot dA. \quad (15)$$

This holds when internal dissipation dominates boundary exchange—i.e. when the system is in a near-equilibrium regime. For cosmological systems this is satisfied on scales $\lesssim 100$ Mpc. For cortical systems it corresponds to the resting-state condition. For RLHF it corresponds to the convergent-training regime (not during rapid phase transitions such as grokking).

6.2 Limitations

1. **B1 parameter-free prediction is an order-of-magnitude estimate.** The cortex scale L_{cortex} enters as L^3 , so α_N is sensitive to this choice. A finer parcellation (e.g. HCP atlas, 180 regions, $L \approx 0.01$ m per parcel) would shift α_N by 10^6 . The prediction P4** should be tested at the parcellation scale, not the whole-brain scale.
2. **B1** replaces a non-optimal observable.** The previous formulation (v0.1) used the Fiedler eigenvalue λ_2 as the structural driver. Empirical data showed $r \approx 0.16$ for λ_2 versus $r \approx -0.97$ for the degree ratio. The revision preserves the cross-domain structure of the EFC framework while correcting the choice of control variable in the neural regime. The unified dynamics (Eq. 2) and Bridge B2 are unaffected.
3. **B2 is exact at the level of equilibrium optima.** The identification Eq. (11) holds for the stationary distributions of both systems. Transient dynamics (gradient descent trajectories, neural spikes) require an additional kinetic theory.
4. T_{cog} **requires independent measurement.** Until T_{cog} is directly extracted from fMRI data, Eq. (13) is a consistency relation, not a standalone prediction.
5. **The bridge does not close the cosmology–RLHF loop directly.** B1 and B2 form a chain (cosmos $\rightarrow$ neural $\rightarrow$ RLHF). A direct cosmology–RLHF bridge would require showing that g_{bar} and β_{KL} are both instances of the same effective temperature concept. This is left for future work.

6.3 Proposed next steps

1. Test P4**: compute η from degree ratio and compare with resting-state MSE estimates from the HCP dataset across multiple subjects.
2. Test P5: use resting-state fMRI $\nabla \dot{S}_{\text{neural}}$ to predict β_{KL}^* for a specific task class; compare with RLHF grid-search optimum on the same task.
3. Derive the kinetic extension of B2 (transient gradient-descent trajectories in RLHF mapped to neural firing dynamics).
4. Formalise the direct cosmology–RLHF bridge via dimensional reduction of Eq. (2) over the configuration space.

7 What Changed from v0.1 to v0.2

For transparency, the following summarises the revision:

Table 3: Summary of changes from v0.1 to v0.2.

Component	v0.1 Status	v0.2 Status
Unified equation (Eq. 2)	Intact	Intact
B1 principle (gradient-driven)	Valid	Valid
B1 implementation (λ_2)	Used Fiedler eigenvalue	Replaced by D_{ratio}
B2 (neural $\rightarrow$ RLHF)	Intact	Intact
Cross-domain predictions	P4*, P5, P6	P4** , P5, P6
DOI	Pending	10.6084/m9.figshare.31940547

The revision from global spectral structure (λ_2) to local degree heterogeneity (D_{ratio}) as the driver of the neural entropy gradient is consistent with the general EFC principle that gradients are governed by proximal, regime-appropriate observables. The previous formulation used a non-optimal observable; the revision corrects this without altering the theoretical structure.

8 Reproducibility

This note presents formal derivations. No original data analysis is included. All parameter values (k , a_G , $g^\dagger$) are taken from Spor 1 [1]; entropy-gradient predictions from Spor 2 [2]; RLHF isomorphism from Spor 3 [3].

References

- [1] Magnusson, M. (2026). Energy-Flow Cosmology: Empirical Validation of the EFC Screening Model Against the Radial Acceleration Relation. Figshare Preprint. [doi:10.6084/m9.figshare.31940469](https://doi.org/10.6084/m9.figshare.31940469)
- [2] Magnusson, M. (2026). EFC-C: A Thermodynamic Framework for Cognitive Entropy and Psychiatric Biomarkers. Figshare Preprint. [doi:10.6084/m9.figshare.31940505](https://doi.org/10.6084/m9.figshare.31940505)
- [3] Magnusson, M. (2026). Reinforcement Learning from Human Feedback as Thermodynamic Entropy Minimisation: A Formal Isomorphism and Testable Predictions. Figshare Preprint. [doi:10.6084/m9.figshare.31940535](https://doi.org/10.6084/m9.figshare.31940535)